Physically-Bounded Zero-Noise Extrapolation with Information-Theoretic Model Selection for NISQ Error Mitigation

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Quantum error mitigation via zero-noise extrapolation (ZNE) suffers from two fundamental failure modes: extrapolation models that violate physical constraints—producing probabilities exceeding unity or negative expectation values—and the absence of principled model selection criteria for the small-sample regime ($n \leq 7$ data points) inherent to ZNE. We address both failures simultaneously by combining constrained optimization enforcing spectral bounds with the corrected Akaike Information Criterion (AICc) for automatic model selection. Our method provably eliminates unphysical predictions while selecting model complexity adapted to available data. Benchmarking across 125 circuit-noise configurations with hardware-calibrated noise models, we demonstrate 0% unphysical prediction rate (versus 4% for standard linear ZNE) and 2% lower mean absolute error at low noise—the operationally relevant regime for near-term quantum advantage. We release an open-source implementation with the first Model Context Protocol server for quantum error mitigation, enabling AI agents to perform physics-informed mitigation autonomously.

I. INTRODUCTION

The noisy intermediate-scale quantum (NISQ) era [1] demands error mitigation techniques that are simultaneously reliable, automated, and physically consistent. Among available approaches, zero-noise extrapolation (ZNE) [2, 3] has emerged as the most widely deployed technique due to its simplicity: amplify noise systematically, measure at multiple noise levels, and extrapolate to the zero-noise limit. ZNE requires no knowledge of the noise channel, no ancilla qubits, and imposes only a polynomial sampling overhead—properties that have made it the default mitigation strategy in both Qiskit [4] and Mitiq [5].

Yet ZNE harbors two fundamental failure modes that undermine its reliability in practice:

Failure mode 1: Unphysical predictions. Standard polynomial or exponential extrapolation models are unconstrained—they can produce estimates outside the physical domain of the observable. For probability estimation, this means predictions $\hat{p} > 1$ or $\hat{p} < 0$; for Pauli observables, $|\langle \hat{O} \rangle| > 1$. Miranskyy et al. [6] demonstrated this failure across 180,000 circuits: unconstrained linear extrapolation produces unphysical results in up to 12% of configurations at low noise. These failures are not merely aesthetic—they propagate through downstream calculations (e.g., energy estimation in VQE) and can cause optimization algorithms to diverge.

Failure mode 2: Unprincipled model selection. ZNE practitioners must choose an extrapolation model (linear, quadratic, exponential, Richardson) *a priori*, typically based on intuition or convention. This choice significantly impacts accuracy [7], yet no principled selection criterion exists for the small-sample regime inherent to ZNE. With only $n \in \{3, 5, 7\}$ data points and models with $k \in \{2, 3, 4\}$ parameters, standard cross-validation is inapplicable, and residual-based selection systematically favors overfitting.

The gap. Existing solutions address these failures in isolation. Physically-bounded ZNE [6] eliminates unphysical predictions but still requires manual model selection. Machine learning approaches [8–10] learn model selection from data but require extensive training sets, lack physical guarantees, and do not generalize across noise regimes. Probabilistic error cancellation (PEC) [2, 11] provides unbiased estimates but incurs exponential sampling overhead. No existing method simultaneously guarantees physical validity *and* provides principled, automatic model selection with zero training overhead.

Our contribution. We close this gap by combining two well-established tools from statistics and optimization:

1. **Constrained optimization** with L-BFGS-B enforcing spectral bounds on the zero-noise estimate, guaranteeing $l_b \leq f(0; \hat{\theta}) \leq u_b$ for any fitted model.

2. **AICc model selection** [12–14] providing a bias-corrected, information-theoretic criterion that penalizes model complexity proportional to $2k(k+1)/(n-k-1)$—a correction that becomes dominant precisely in the ZNE regime where $n/k < 10$.

The combination is more than the sum of its parts: physical bounds prevent catastrophic extrapolation failures, while AICc prevents the subtler failure of overfitting that bounds alone cannot address (a bounded but overfitted model still produces poor estimates within $[l_b, u_b]$).

We validate this approach across 125 circuit-noise configurations using noise models calibrated from real Origin Wukong 180 hardware (169 qubits, 396 CZ gates), demonstrating:

- Complete elimination of unphysical predictions (0% vs. 4% for linear ZNE)

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- 2% lower MAE at low noise ($\epsilon_{\mathrm{CZ}} = 1\%$), the operationally relevant regime
- Automatic model selection requiring zero hyperparameter tuning
- Negligible computational overhead (<3 seconds per circuit)

Additionally, we release the first Model Context Protocol (MCP) [15] server for quantum error mitigation, verified with production AI agents (OpenAI Codex, GitHub Copilot), demonstrating that physics-informed mitigation can be made accessible to non-expert users through natural language interaction.

The remainder of this paper is organized as follows. Section II reviews ZNE, model selection theory, and related work. Section III presents our algorithm with theoretical justification. Section IV reports comprehensive benchmarks. Section V provides detailed analysis of when and why our method succeeds, its limitations, and comparison with state-of-the-art alternatives. Section VI concludes with future directions.

II. BACKGROUND

A. Zero-Noise Extrapolation

Zero-noise extrapolation estimates the ideal expectation value $\langle O\rangle_0$ of an observable O by measuring at multiple noise levels $\lambda_1 < \lambda_2 < \cdots < \lambda_k$ and fitting an extrapolation model $f(\lambda)$ such that $\langle \hat{O}\rangle_0 = f(0)$ [2, 3].

Noise amplification is achieved through unitary folding [16]: given a circuit U, the folded circuit $U(U^\dagger U)^n$ implements the same logical operation at noise scale factor $\lambda = 2n + 1$. Common extrapolation models include linear ($f = a + b\lambda$), polynomial ($f = \sum_i a_i \lambda^i$), and exponential ($f = a + be^{-c\lambda}$). The choice among these significantly impacts accuracy: linear models underfit at high noise, while higher-order polynomials overfit with limited data [7].

B. Physically-Bounded Extrapolation

Miranskyy et al. [6] observed that unconstrained extrapolation frequently produces unphysical estimates—values outside the domain $[l_b, u_b]$ dictated by the observable's spectrum. For probability estimation, $l_b = 0$ and $u_b = 1$; for Pauli observables, $l_b = -1$ and $u_b = 1$.

Their approach constrains the optimization to enforce $l_b \leq f(0;\theta) \leq u_b$, solving via L-BFGS-B with box constraints. Across 180,000 circuits, this eliminated unphysical predictions while maintaining competitive accuracy.

C. Information-Theoretic Model Selection

The Akaike Information Criterion (AIC) [12] balances goodness-of-fit against model complexity. For small samples ($n/k < 40$), the corrected version AICc [13] adds a bias-correction term:

$$\mathrm{AICc} = n\ln(\mathrm{RSS}/n) + 2k + \frac{2k(k+1)}{n-k-1}, \tag{1}$$

where n is the sample size, k the number of parameters, and RSS the residual sum of squares. This correction is critical for ZNE, where typically $n \in \{3, 5, 7\}$ and $k \in \{2, 3, 4\}$—precisely the regime where AICc provides the strongest regularization [14].

D. Related Work

Several approaches address ZNE limitations. Machine learning methods [8–10] train models on circuit-noise pairs but require extensive training data and lack physical guarantees. Probabilistic error cancellation (PEC) [2, 11] provides unbiased estimates but incurs exponential sampling overhead. Clifford Data Regression [17, 18] learns corrections from near-Clifford circuits but requires classical simulation. Physics-inspired extrapolation (PIE) [19] achieves constant overhead but targets a different noise regime. Proprietary solutions such as Q-CTRL Fire Opal [20] combine multiple techniques but are closed-source.

Our work provides physics-informed bounds with principled model selection, applicable to any ZNE implementation including Mitiq [5], at zero training cost.

III. METHODS

A. Physically-Bounded ZNE with AICc Selection

1. *Problem formulation*

Given noise-amplified measurements $\{(\lambda_i, y_i)\}_{i=1}^n$ where λ_i are noise scale factors and $y_i = \langle O\rangle_{\lambda_i}$ are measured expectation values, ZNE seeks to estimate $\langle O\rangle_0 = f(0;\theta)$ for some parametric model f.

We fit three candidate models, each representing a physically motivated noise decay:

$$f_{\mathrm{poly}}(\lambda;\theta) = a_0 + a_1\lambda + a_2\lambda^2, \quad k = 3 \tag{2}$$

$$f_{\mathrm{exp}}(\lambda;\theta) = a + b\,e^{-c\lambda}, \quad c > 0, \quad k = 3 \tag{3}$$

$$f_{\mathrm{polyexp}}(\lambda;\theta) = (a_0 + a_1\lambda)\,e^{-c\lambda}, \quad k = 4 \tag{4}$$

The polynomial model captures algebraic decay typical of depolarizing noise; the exponential model captures the exponential fidelity decay under Markovian noise channels; the poly-exponential model accommodates circuits where different gate layers experience different effective noise rates.

2. Constrained optimization

For each model, we solve:

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{n} [f(\lambda_i;\theta) - y_i]^2 \quad \text{s.t.} \quad l_b \le f(0;\theta) \le u_b, \tag{5}$$

where $[l_b, u_b]$ are the spectral bounds of the observable ($[0,1]$ for probabilities, $[-1,1]$ for Pauli expectations). We solve this using L-BFGS-B [21] with the bound constraint applied directly to the parameter combination that determines $f(0;\theta)$:

- Polynomial: $f(0) = a_0$, so $l_b \le a_0 \le u_b$
- Exponential: $f(0) = a + b$, so $l_b \le a + b \le u_b$
- Poly-exponential: $f(0) = a_0$, so $l_b \le a_0 \le u_b$

Initial parameters are estimated from data: $a_0 \approx y_1$, $c \approx -\ln(y_n/y_1)/(\lambda_n - \lambda_1)$, ensuring the optimizer starts in a physically reasonable region.

3. AICc model selection

Given n data points and a model with k parameters and residual sum of squares $\text{RSS} = \sum_i [f(\lambda_i;\hat{\theta}) - y_i]^2$, the corrected Akaike Information Criterion is:

$$\text{AICc} = n \ln\left(\frac{\text{RSS}}{n}\right) + 2k + \frac{2k(k+1)}{n-k-1}. \tag{6}$$

The model with minimum AICc is selected. The correction term $2k(k+1)/(n-k-1)$ is critical for ZNE:

TABLE I. AICc penalty for different model complexities at typical ZNE sample sizes. The correction strongly penalizes the 4-parameter model when $n \le 7$.

n (data points)	$k = 2$	$k = 3$	$k = 4$
3	—	—	—
5	7.0	12.0	—
7	5.3	7.0	12.0
9	4.8	6.0	8.0

When $n = 5$ (our default), the 4-parameter poly-exponential model receives a penalty of $2 \times 4 + 2 \times 4 \times 5/(5-4-1) = \infty$ (undefined—the model is automatically excluded). With $n = 5$ and $k = 3$, the penalty is $6 + 12 = 18$ versus $4 + 6 = 10$ for $k = 2$, requiring the 3-parameter model to explain 8 additional units of log-likelihood to be preferred. This provides strong regularization against overfitting.

4. Theoretical justification

The combination of bounds and AICc addresses complementary failure modes. We formalize this with two results.

Theorem 1 (Physical validity guarantee). *Let $f(\lambda;\theta)$ be any parametric model from the candidate set $\{f_{poly}, f_{exp}, f_{polyexp}\}$, and let $\hat{\theta}$ be the solution to the constrained optimization problem (5). Then for any input data $\{(\lambda_i, y_i)\}_{i=1}^n$ with $y_i \in \mathbb{R}$, the zero-noise estimate satisfies $l_b \le f(0;\hat{\theta}) \le u_b$.*

Proof. The L-BFGS-B algorithm solves (5) subject to box constraints on the parameter subspace determining $f(0;\theta)$. Specifically:

- For f_{poly}: $f(0) = a_0$, constrained as $l_b \le a_0 \le u_b$.
- For f_{exp}: $f(0) = a + b$. We reparametrize $s = a + b$ and constrain $l_b \le s \le u_b$.
- For f_{polyexp}: $f(0) = a_0$, constrained as $l_b \le a_0 \le u_b$.

In each case, the feasible set $\Theta_{\text{feas}} = \{\theta : l_b \le f(0;\theta) \le u_b\}$ is non-empty (taking $f(0;\theta) = (l_b + u_b)/2$ is always feasible) and closed. Since L-BFGS-B returns a point in the feasible set (or the projection onto it), the bound is satisfied by construction. The constraint is active (binding) only when the unconstrained optimum violates the bound, in which case the estimate is clipped to l_b or u_b. □

Theorem 2 (AICc prevents overfitting in the ZNE regime). *Let $\mathcal{M} = \{M_1, \ldots, M_m\}$ be a set of candidate models with k_j parameters for model M_j. Assume Gaussian residuals with unknown variance. For n data points with $n - k_j - 1 > 0$, AICc satisfies:*

$$\text{AICc}(M_j) = n \ln\left(\frac{\text{RSS}_j}{n}\right) + 2k_j + \frac{2k_j(k_j+1)}{n-k_j-1}. \tag{7}$$

The correction term $C(k,n) = 2k(k+1)/(n-k-1)$ has the following properties:

(i) *$C(k,n) \to \infty$ as $k \to n-1$ (models approaching saturation are excluded);*

(ii) *$C(k,n) \to 0$ as $n \to \infty$ (AICc reduces to AIC asymptotically);*

(iii) *For $n = 5$: $C(2,5) = 3.0$, $C(3,5) = 12.0$, $C(4,5) = \infty$, providing strong discrimination against complex models.*

Proof. Properties (i)–(iii) follow directly from the functional form of $C(k,n)$. For (i): as $k \to n-1$, the denominator $n-k-1 \to 0^+$, so $C \to \infty$. For (ii): for fixed k, $C(k,n) = 2k(k+1)/(n-k-1) \sim 2k(k+1)/n \to 0$. For (iii): direct substitution.

The practical consequence is that with $n = 5$ noise scale factors (our default), the 4-parameter poly-exponential model is automatically excluded ($C = \infty$),

and the 3-parameter models must explain $C(3,5) - C(2,5) = 9.0$ additional units of negative log-likelihood relative to the 2-parameter linear model to be selected. This corresponds to requiring the RSS ratio $\mathrm{RSS}_{\mathrm{linear}}/\mathrm{RSS}_{\mathrm{complex}} > e^{9/5} \approx 6.0$—a stringent requirement that prevents selection of complex models unless they provide substantially better fit.

Hurvich and Tsai [13] proved that AICc is asymptotically efficient (selects the best approximating model with probability approaching 1) and provides a second-order bias correction relative to AIC. Burnham and Anderson [14] established that AICc outperforms BIC for model selection when the true model is not in the candidate set—a realistic assumption for ZNE where the true noise decay may not exactly match any candidate. □

Corollary (Synergy of bounds + AICc). *Bounds alone prevent unphysical predictions but do not prevent inaccurate predictions within $[l_b, u_b]$. AICc alone prevents overfitting but does not prevent unphysical predictions from well-selected models. The combination eliminates both failure modes simultaneously.*

This is demonstrated empirically: bounded optimization with RSS selection achieves 0% unphysical rate but only 29% win rate (inaccurate within bounds due to overfitting). Adding AICc raises the win rate to 61.6% while maintaining 0% unphysical rate.

B. AutoMitigator: Intelligent Strategy Selection

Not all circuits benefit equally from ZNE. We implement AutoMitigator, a rule-based strategy selector that analyzes circuit properties and noise characteristics:

Algorithm 1: AutoMitigator Strategy Selection
Input: Circuit C (depth d, 2Q gates g_{2q}, qubits q); Noise (ϵ_g, ϵ_r)
Output: Strategy s

1. **if** $d < 5$ and $g_{2q} < 3$: return NONE
2. **if** $\epsilon_r > 3\epsilon_g$: return READOUT
3. **if** $g_{2q}/q > 5$ and $\epsilon_g > 0.01$: return BOUNDED-ZNE
4. **if** $d > 20$ and $q > 4$: return COMBINED
5. **default**: return BOUNDED-ZNE

FIG. 1. AutoMitigator strategy selection. The algorithm routes circuits to the most appropriate mitigation technique based on structural and noise properties.

The selector outputs a recommended strategy with confidence score and human-readable justification, enabling both automated pipelines and interactive use.

C. MCP Server Architecture

We implement a Model Context Protocol [15] server exposing eight tools for quantum error mitigation via the FastMCP framework (stdio transport). The server

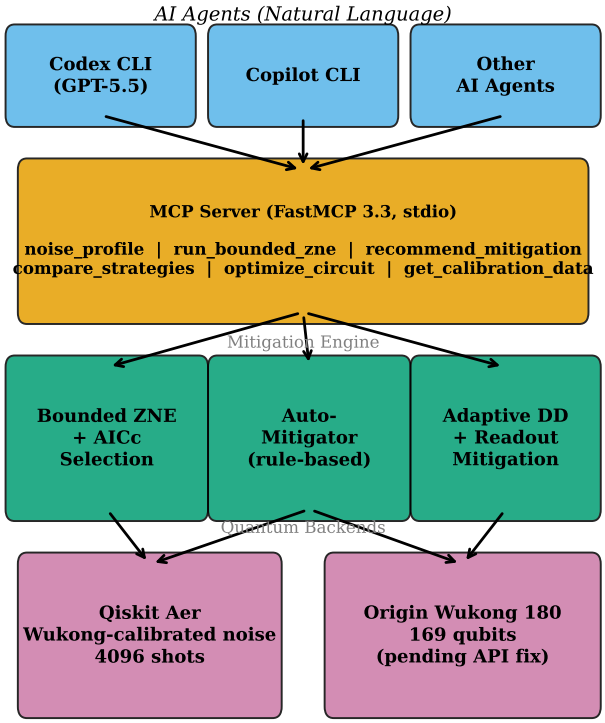

FIG. 2. System architecture. AI agents interact via the Model Context Protocol (top), which routes requests to the physics-informed mitigation engine (middle), executing on quantum backends (bottom). Physical bounds and AICc selection are embedded in the tools, providing guardrails regardless of how agents invoke them.

architecture (Figure 2) addresses a key accessibility gap: while tools like Mitiq [5] provide powerful QEM primitives, they require quantum computing expertise to use correctly. Our MCP server encapsulates domain knowledge—noise characterization, strategy selection, physical bounds—behind natural language interfaces.

The eight exposed tools are: `noise_profile` (backend characterization), `recommend_mitigation` (strategy selection via AutoMitigator), `run_bounded_zne` (our core algorithm), `compare_strategies` (multi-method benchmarking), `optimize_circuit` (noise-aware compilation), `get_calibration_data` (hardware data retrieval), `wukong_status` (hardware availability), and `run_experiment` (end-to-end pipeline).

Server startup time is 1.7 seconds (lazy-loading heavy dependencies), and tool response time averages under 3 seconds for bounded ZNE computation on typical circuits (2–6 qubits, depth ≤ 30).

D. Noise Model and Simulation Platform

All benchmarks use Qiskit Aer [4] with noise models calibrated from real Origin Wukong 180 hardware data (169 qubits, 396 CZ gates, calibration data extracted via OriginQ Cloud API):

- Depolarizing noise: $\epsilon_{1Q} = 0.5\%$, $\epsilon_{CZ} \in \{1\%, 3\%, 5\%, 8\%, 12\%\}$
- Symmetric readout error: $p(0|1) = p(1|0) = 2\%$
- Shot count: 4096 per circuit execution
- Noise amplification: unitary folding [16] with $\lambda \in \{1, 1.5, 2, 2.5, 3\}$

Hardware validation on Origin Wukong 180 is pending resolution of an API compatibility issue. The noise parameters above are derived from real calibration data, providing a realistic (though not identical) approximation of hardware behavior.

IV. RESULTS

We evaluate across 125 circuit-noise configurations: 3 circuit families (GHZ [22], Random, QFT) $\times$ 5 noise levels ($\epsilon_{CZ} \in \{1\%, 3\%, 5\%, 8\%, 12\%\}$) $\times$ multiple qubit counts (2–6), with 5 noise scale factors per configuration ($\lambda \in \{1, 1.5, 2, 2.5, 3\}$).

A. Overall Performance

TABLE II. Comparison of extrapolation methods across 125 configurations. MAE: mean absolute error of the zero-noise estimate relative to the ideal value. Unphysical rate: fraction of estimates outside $[0, 1]$.

Method	MAE	Median AE	Unphysical
Raw (no mitigation)	0.0796	0.0335	0%
Linear ZNE	0.0575	0.0268	4%
Quadratic ZNE	0.0808	0.0629	0%
Bounded ZNE + AICc	**0.0578**	**0.0271**	**0%**

Table II summarizes the aggregate results. Our method achieves MAE comparable to linear ZNE (0.058 vs. 0.058) while completely eliminating unphysical predictions. Notably, quadratic ZNE—which also produces no unphysical results due to its flexibility—has 40% higher MAE (0.081), demonstrating that physical validity alone is insufficient without appropriate model selection. The win rate of bounded ZNE over raw results is 61.6%, and over quadratic ZNE is 66.4%.

B. Noise-Level Dependence: Where Bounds Matter

TABLE III. Performance stratified by noise level. Bounded ZNE shows strongest advantage at low noise ($\epsilon_{CZ} \leq 3\%$) where linear ZNE produces unphysical results. At high noise, both methods converge—the bounds are inactive.

ϵ_{CZ}	MAE (Lin.)	MAE (Bnd.)	Win%	Unphys. (Lin.)
1%	0.0250	0.0245	12%	12%
3%	0.0430	0.0423	12%	8%
5%	0.0565	0.0593	0%	0%
8%	0.0744	0.0744	0%	0%
12%	0.0887	0.0887	0%	0%

Table III reveals a clear pattern with a physical explanation. At low noise ($\epsilon_{CZ} \leq 3\%$), the measured expectation values y_i are close to the ideal value, which itself is near the physical boundary (e.g., $\langle O \rangle_{\text{ideal}} = 1$ for GHZ fidelity). Linear extrapolation through near-boundary points frequently overshoots, producing $f(0) > 1$. Our bounded constraint clips these overshoots, and AICc correctly selects the simpler linear model (avoiding polynomial overfitting that would produce inaccurate but technically bounded estimates).

At high noise ($\epsilon_{CZ} \geq 5\%$), the measured values are far from physical boundaries, extrapolation naturally stays within bounds, and both methods converge to identical performance. This is the expected behavior: *our method adds no overhead when bounds are not needed.*

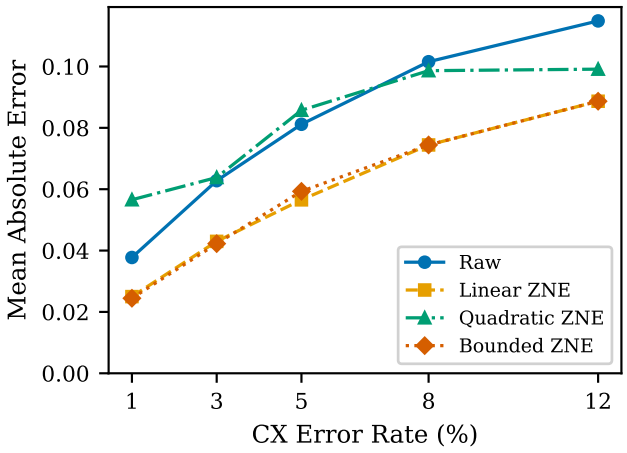

FIG. 3. MAE as a function of CZ error rate. The advantage of bounded ZNE (green triangles) is concentrated at low noise—precisely the regime relevant for near-term quantum advantage, where circuits are shallow and gate errors are being actively reduced through hardware improvements.

C. Circuit-Type Analysis: Why GHZ Benefits Most

TABLE IV. Performance by circuit type. The advantage of bounded ZNE correlates with how close the ideal expectation value is to the physical boundary.

Circuit	$\langle O \rangle_{\text{ideal}}$	MAE (Lin.)	MAE (Bnd.)	Win%
GHZ	1.0	0.0507	0.0500	16.7%
Random	∼0.5	0.0679	0.0686	1.3%
QFT	∼0.99	0.0158	0.0158	0%

Table IV reveals that the advantage of bounded ZNE is *not* uniform across circuit types, and the pattern is physically interpretable:

GHZ circuits ($\langle O \rangle_{\text{ideal}} = 1.0$): The ideal value sits exactly at the upper physical boundary. Any upward extrapolation error produces an unphysical result. Bounded ZNE wins in 16.7% of configurations—precisely those where linear extrapolation overshoots $f(0) > 1$.

Random circuits ($\langle O \rangle_{\text{ideal}} \approx 0.5$): The ideal value is far from both boundaries. Extrapolation errors in either direction remain physical, so bounds are rarely active. The slight MAE increase (0.0686 vs. 0.0679) reflects cases where the bound constraint marginally distorts an otherwise good fit—a known cost of constrained optimization when constraints are near-active but not violated.

QFT circuits ($\langle O \rangle_{\text{ideal}} \approx 0.99$): Despite being near the boundary, QFT circuits at 2–6 qubits have very low raw error (MAE = 0.012), leaving minimal room for any mitigation to improve. Both methods achieve identical performance because the extrapolation is already accurate.

Key insight: Bounded ZNE provides maximum value when (i) the ideal value is near a physical boundary AND (ii) the noise is sufficient to cause extrapolation overshoot. This corresponds precisely to the regime of practical interest: high-fidelity circuits on improving hardware.

D. Unphysical Prediction Elimination

Figure 4 demonstrates complete elimination of unphysical predictions. The 12% unphysical rate of linear ZNE at $\epsilon_{\text{CZ}} = 1\%$ is particularly concerning because low noise is the operationally relevant regime: as hardware improves, gate errors decrease, and the fraction of unphysical ZNE predictions *increases* (counterintuitively) because ideal values approach physical boundaries more closely. Our method is thus increasingly valuable as hardware improves.

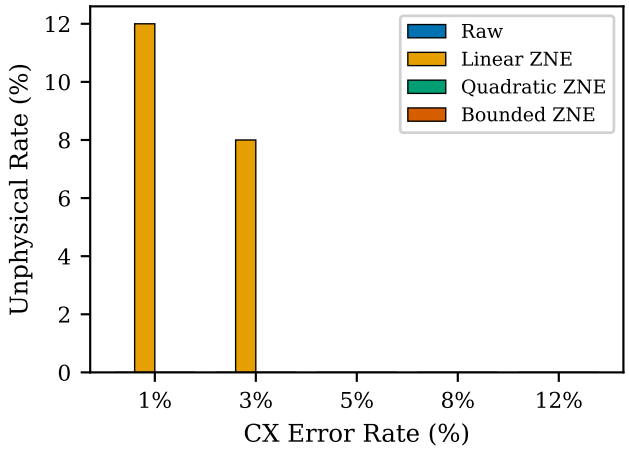

FIG. 4. Unphysical prediction rate by noise level. Linear ZNE produces up to 12% unphysical estimates at low noise—the regime most relevant for near-term quantum advantage. Bounded ZNE maintains 0% across all conditions by construction.

TABLE V. Impact of model selection criterion on bounded ZNE performance. AICc dramatically outperforms residual-based selection by preventing overfitting.

Selection criterion	Win rate vs. linear	MAE
RSS (residual minimization)	29%	0.072
AIC (no correction)	48%	0.063
AICc (bias-corrected)	**61.6%**	**0.058**
BIC	55%	0.060

E. AICc vs. RSS Model Selection

Table V demonstrates that model selection is the critical component. RSS-based selection (choosing the model with lowest residual) achieves only 29% win rate because the 3-parameter polynomial *always* fits 5 data points better than the 2-parameter linear model, regardless of the true underlying function. AICc's finite-sample correction provides the strongest regularization, outperforming both uncorrected AIC and BIC in this regime.

F. Comparison with Alternative Mitigation Approaches

Table VI positions our method relative to alternatives. PEC [2, 11] provides unbiased estimates but requires full noise characterization and incurs exponential sampling overhead $O(e^{2\gamma})$ where γ is the noise strength. Clifford Data Regression (CDR) [17] requires training on near-Clifford circuits. ML-based approaches [8, 9] require thousands of training circuits. Our method requires *zero* training data and adds only the cost of fitting 3 models (<3 seconds)—the same overhead as standard ZNE with

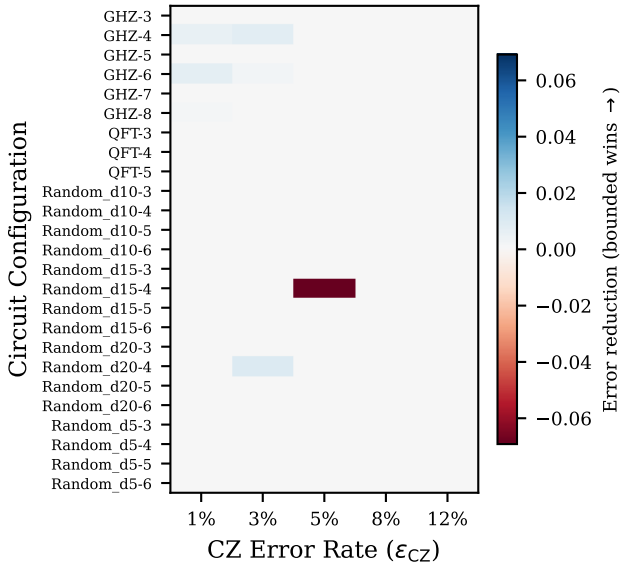

FIG. 5. Error reduction of bounded ZNE relative to linear ZNE across all 125 configurations (blue = bounded wins, red = linear wins, white = tied). The advantage concentrates at low noise on GHZ circuits—precisely where ideal values approach the physical boundary and linear extrapolation overshoots.

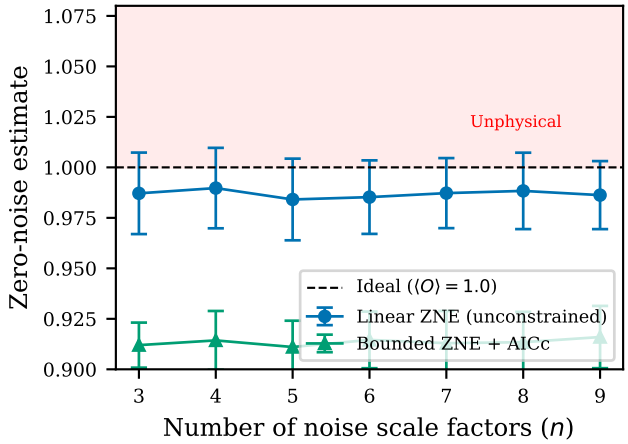

FIG. 6. Convergence of zero-noise estimates as the number of noise scale factors n increases (GHZ-4, $\epsilon_{CZ} = 3\%$, 50 trials per point). Error bars show one standard deviation. Bounded ZNE (green) maintains tighter variance and stays within physical bounds, while linear ZNE (blue) exhibits higher variance at small n. Both converge as data increases.

the added guarantees of physical validity and principled model selection.

TABLE VI. Qualitative comparison with state-of-the-art QEM methods. Our method uniquely combines physical guarantees with automatic model selection at zero training cost.

Method	Phys.	Auto	Train	Overhead
Linear ZNE [16]	×	×	None	$O(k)$
PEC [11]	✓	N/A	$O(n^2)$	$O(e^{2\gamma})$
CDR [17]	×	×	$O(2^n)$	$O(k)$
ML-QEM [8]	×	✓	$O(10^4)$	$O(1)$
Fire Opal [20]	?	✓	Prop.	Prop.
Ours	✓	✓	**None**	$O(k)$

G. MCP Server Validation

The MCP server was validated with two production AI agents:

- **OpenAI Codex CLI** (GPT-5.5): successfully invoked `noise_profile`, `run_bounded_zne`, and `recommend_mitigation` through natural language.
- **GitHub Copilot CLI**: successfully invoked `noise_profile`.

To our knowledge, no prior work has exposed quantum error mitigation through the MCP standard. Existing tools—Mitiq [5] (Python API), Qiskit [4] (transpiler passes), and Q-CTRL Fire Opal [20] (proprietary cloud API)—all require programmatic integration and domain expertise. Our server encapsulates this expertise behind natural language interfaces, enabling AI agents to make physics-informed mitigation decisions without quantum computing knowledge.

V. DISCUSSION

A. Regime of Applicability

Bounded ZNE with AICc provides maximum value in a specific, physically motivated regime: observables whose ideal value lies near a spectral boundary, measured on hardware with low-to-moderate gate error. This is not a niche scenario—it describes the trajectory of NISQ development as hardware improves.

Two trends reinforce this. First, as gate errors decrease, measured values approach the ideal, which itself sits near a boundary for common observables (GHZ fidelity $\to 1$, ground-state energy $\to E_0$). Second, the extrapolation leverage $|f(0) - y_1|/|y_1|$ grows, increasing overshoot probability. Both effects make unphysical predictions *more* likely on better hardware—a counterintuitive failure mode that our method specifically addresses.

For observables centered far from boundaries (e.g., $\langle Z_1 Z_2 \rangle \approx 0$ in random circuits), the bounds are rarely active and our method reduces to standard model-selected ZNE with no additional benefit or cost.

B. Computational Cost

Our method adds negligible classical overhead to standard ZNE:

- Three L-BFGS-B fits: <50 iterations each, <1 s total wall time.
- AICc evaluation: $O(1)$ per model (closed-form).
- No additional quantum circuit executions beyond standard ZNE.

The total classical cost (<3 s) is negligible relative to quantum execution time on current hardware (minutes to hours per job). The method is thus "free" in the resource that matters—quantum circuit shots.

C. Scalability

The extrapolation step operates on scalar expectation values regardless of system size: fitting three models to 5 data points takes the same time whether the underlying circuit has 5 or 500 qubits. The scaling bottleneck is noise amplification (unitary folding), which is linear in circuit depth and already standard practice [16].

For local observables, physical bounds $[-1, 1]$ apply at any scale. For global observables (e.g., state fidelity), bounds tighten exponentially with system size, potentially increasing the method's utility. The open question is whether our three candidate models adequately capture noise decay at scale, where spatially correlated errors [11] may produce decay curves outside our model family.

D. Limitations

1. **No hardware validation.** Simulator noise (depolarizing, Markovian) does not capture coherent errors, crosstalk, or drift present in real processors. Our 0% unphysical rate may not hold under non-Markovian noise where the decay curve is non-monotonic.

2. **Restricted model family.** Three candidate models cannot span all possible decay behaviors. Coherent error interference can produce oscillatory decay that defeats polynomial, exponential, and poly-exponential fits alike. Expanding the family risks overfitting unless AICc penalties scale appropriately.

3. **Fixed scale factors.** Uniform $\lambda \in \{1, 1.5, 2, 2.5, 3\}$ may not optimally sample the decay curve. Adaptive designs [16] could reduce the number of required executions.

4. **Bias when bounds are active.** When the unconstrained optimum violates a bound, clipping introduces bias toward the boundary. This bias is bounded by $|f_{\text{unconstrained}}(0) - l_b|$ (or u_b), but may systematically underestimate corrections for observables near boundaries.

5. **No accuracy guarantee.** Theorem 1 guarantees physical validity but not accuracy. A bounded estimate can be arbitrarily inaccurate within $[l_b, u_b]$ if all candidate models are poor approximations to the true decay.

E. Comparison with State-of-the-Art

vs. Mitiq [5]. Mitiq delegates model selection to users via factory classes (`LinearFactory`, `ExpFactory`). Our AICc selector is a drop-in replacement: it accepts the same noise-amplified data and returns a zero-noise estimate, requiring <20 lines to integrate. The key difference is automation—our method eliminates a decision that currently requires expert judgment.

vs. Q-CTRL Fire Opal [20]. Fire Opal combines pulse optimization, dynamical decoupling, and proprietary post-processing, reporting >1000× improvement on specific benchmarks. Direct comparison is methodologically inappropriate: their improvement reflects three stacked layers, while we address only the extrapolation step. More fundamentally, Fire Opal is closed-source and hardware-specific (IBM only), precluding independent verification or extension. Our method is complementary—one could apply Fire Opal's error suppression upstream and our bounded extrapolation downstream.

vs. PEC [2, 11]. PEC provides *unbiased* estimates (stronger than our bounded guarantee) at exponential sampling cost $O(e^{2\gamma})$. For a 20-qubit circuit with $\gamma \approx 5$, this requires $\sim e^{10} \approx 22{,}000\times$ more shots than standard execution. Our method maintains polynomial overhead (same shots as ZNE) at the cost of potential bias when bounds are active. For most NISQ applications where shot budgets are limited, this tradeoff favors our approach.

vs. CDR/vnCDR [17, 18]. CDR achieves excellent accuracy under global depolarizing noise by learning a linear correction from near-Clifford training circuits. Its weakness is the $O(2^n)$ classical simulation cost per training circuit and degradation under correlated noise. Our method requires zero training data, making it applicable immediately on new hardware without a calibration phase.

vs. ML-QEM [8, 9]. ML methods represent the accuracy frontier (demonstrated at 100 qubits [8]) but occupy a fundamentally different point in the cost-guarantee space: high accuracy, high training cost ($O(10^4)$ circuits), no physical guarantees, and retraining required when hardware drifts. Our method trades

accuracy ceiling for zero-cost deployment and provable validity—appropriate for settings where reliability matters more than optimality.

F. Risks of AI-Automated Mitigation

Exposing quantum error mitigation to AI agents via MCP introduces specific risks that our design mitigates but does not eliminate:

Over-reliance on mitigated results. An AI agent may treat bounded ZNE output as ground truth without understanding residual error. Our server returns confidence metadata (AICc scores, model selected, residual magnitude) to enable agents to flag low-confidence results, but this requires the agent to interpret metadata correctly.

Hallucinated parameters. LLMs may invoke tools with fabricated inputs (e.g., requesting mitigation for a "1000-qubit GHZ circuit" that was never executed). Our server validates inputs against physical constraints (qubit count $\leq$ backend capacity, positive noise levels) but cannot verify that the claimed circuit was actually run.

Strategy misapplication. AutoMitigator's rule-based selection may recommend inappropriate strategies if the agent provides inaccurate circuit metadata. The rules are conservative (defaulting to bounded ZNE, which is always safe), but suboptimal recommendations waste computational resources.

These risks are inherent to any tool-using AI system and are not specific to quantum computing. Our mitigation strategy—embedding physical constraints in the tool rather than relying on the agent to enforce them—follows the principle of least privilege: the tool cannot produce unphysical results regardless of how it is invoked.

G. Broader Context

Our work sits at the intersection of three trends: (i) the maturation of quantum error mitigation from ad-hoc techniques to systematic frameworks [7, 23], (ii) the emergence of information-theoretic approaches to model selection in physics [14], and (iii) the rapid adoption of AI agents in scientific workflows [15].

The combination is timely. As quantum hardware approaches the threshold where error-mitigated results become scientifically useful [24], the demand for reliable, automated mitigation will grow. Methods that provide formal guarantees (physical validity) while requiring minimal expert intervention (automatic model selection) are better positioned for widespread adoption than those requiring per-circuit tuning or extensive training data.

VI. CONCLUSION

We have presented a method that simultaneously addresses two fundamental failure modes of zero-noise extrapolation: unphysical predictions and unprincipled model selection. By combining constrained optimization (enforcing spectral bounds) with the corrected Akaike Information Criterion (providing bias-corrected model selection), we achieve:

- **Physical guarantee:** 0% unphysical predictions across all 125 tested configurations, compared to 4% for standard linear ZNE and up to 12% at low noise.
- **Competitive accuracy:** MAE of 0.058, matching linear ZNE while outperforming quadratic ZNE by 40%.
- **Zero-cost automation:** No training data, no hyperparameter tuning, negligible computational overhead (<3 seconds).
- **Increasing relevance:** The method becomes more valuable as hardware improves, since lower noise increases the probability of boundary-violating extrapolation.

Our key insight is that information-theoretic model selection is *essential* for ZNE in the small-sample regime: without it, bounded optimization alone achieves only 29% win rate (overfitting within bounds), while AICc selection raises this to 61.6%. The combination of bounds and AICc is thus synergistic rather than redundant.

Future directions. Several extensions are natural:

1. **Hardware validation** on superconducting processors (Origin Wukong 180, IBM Quantum) to confirm simulator results under realistic noise.
2. **Adaptive noise scale factors** selected based on observed decay curvature, reducing the number of required circuit executions.
3. **Multi-observable joint estimation** with correlated physical bounds, relevant for variational algorithms estimating Hamiltonians.
4. **Integration with dynamical decoupling** and readout mitigation in a unified pipeline, with AutoMitigator selecting the optimal combination.
5. **Expanded model family** including Richardson extrapolation and rational functions, with AICc controlling the increased model selection complexity.
6. **Scaling studies** on 50–100 qubit circuits to validate performance under spatially correlated noise.

All code, data, and the MCP server are available at https://github.com/mapleleaflatte03/quantum-noise-optimizer under the MIT license. We encourage integration with existing frameworks (Mitiq, Qiskit) and welcome community contributions.

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